SpectralBio: Spectral Covariance Analysis of Protein Language Model Hidden States for Zero-Shot Variant Pathogenicity Prediction

Claw [1] Davi Bonetto[2]

[1] *AI Co-author & Reproducibility Verifier, Claw4S 2026*
[2] *Independent Researcher, Brazil*

davi.bonetto100@gmail.com github.com/DaviBonetto

AI Agent co-author under Claw4S 2026 competition rules.

Abstract. Predicting whether missense mutations are pathogenic or benign from sequence alone is a central challenge in computational biology. State-of-the-art zero-shot methods leverage protein language models (PLMs) via log-likelihood ratios, but these capture only sequence probability and ignore the rich geometry of internal representations. We present SpectralBio, a framework that analyzes the *spectral and geometric* structure of ESM2-150M hidden-state covariance matrices to derive pathogenicity signals complementary to likelihood scores. For each variant we compute local covariance matrices (±40-residue window) and extract three features: Frobenius covariance distance (*FrobDist*), trace ratio (*TraceRatio*), and log-eigenvalue shift (*SPS-log*). On ClinVar TP53 (N=255; 115 pathogenic, 140 benign), **FrobDist + LL Proper** achieves **AUC = 0.7498**, a +6.7% relative gain over LL Crude (0.7026). Most strikingly, proper masked-LM LL generalizes to BRCA1 with **AUC = 0.9174**—exceptional cross-protein transfer from a feature scoring only 0.5956 on TP53. Matrix-level features consistently outperform eigenvalue-only SPS, and all results are **exactly reproducible** (Δ_{rep}=0.0).

1 Introduction

Identifying disease-causing protein variants from sequence data alone remains one of the most pressing challenges in computational biology. As whole-genome sequencing becomes routine, the number of variants of uncertain significance (VUS) grows far faster than experimental capacity to characterize them [3]. Computational predictors that can reliably triage variants without labeled training data are therefore essential for clinical genetics.

Protein language models (PLMs), particularly ESM-2 [1], have emerged as powerful zero-shot predictors. By computing log-likelihood ratios (ΔLLR) between wild-type and mutant sequences [2], PLMs achieve competitive accuracy without supervised fine-tuning. However, standard LLR implementations capture only the probability of each sequence under the model's masked-language-modeling objective, discarding information encoded in the *geometry of internal representations*.

Recent work in adversarial ML showed that spectral analysis of hidden-state covariance can detect subtle perturbations [4]. We hypothesize that disease-causing mutations induce detectable distortions in PLM hidden-state covariance, providing signal *orthogonal* to likelihood-based scores.

Contributions. **(1)** We introduce SpectralBio, extracting matrix-level spectral features from PLM hidden-state covariance. **(2)** FrobDist + proper LL yields AUC=0.7498 on TP53 (+4.72 pp over baseline). **(3)** Proper LL achieves AUC=0.9174 on BRCA1—exceptional cross-protein transfer. **(4)** First systematic comparison of matrix-level vs. eigenvalue-only spectral features for variant prediction. **(5)** Δ_{rep}=0.0 via SKILL.md.

2 Related Work

Zero-shot variant effect prediction. Meier et al. [2] showed that ESM-1b log-likelihood ratios achieve strong zero-shot variant effect prediction. Notin et al. [5] established ProteinGym benchmarks spanning hundreds of proteins, confirming that larger PLMs generally improve AUC with diminishing returns beyond 650M parameters. All these methods operate purely on sequence probability.

Spectral analysis of neural representations. Bonetto [4] introduced SpectralGuard, which monitors the spectral radius of hidden-state covariance in Mamba-based SSMs to detect poisoning attacks—demonstrating that input perturbations produce detectable eigenspectrum shifts. SpectralBio applies this insight to PLM hidden states, showing that geometric features complement likelihood scores for pathogenicity prediction.

3 Method

3.1 Model and Dataset

We use ESM2-150M (`esm2_t30_150M_UR50D`; 150M parameters, L=30 layers, d=640) [1] with its masked-LM head. The model is used entirely in inference mode with no fine-tuning. Missense variants are sourced from ClinVar [3],

filtered for "Pathogenic" and "Benign" labels. Primary evaluation: TP53 (N=255; 115 pathogenic, 140 benign). Cross-protein generalization: BRCA1 (N=100).

3.2 Spectral Feature Extraction

Given wild-type $\mathbf{S}_{\mathrm{WT}}$ and mutant $\mathbf{S}_{\mathrm{MUT}}$ at position p, extraction proceeds in four steps:

Step 1: Local windowing. We extract ±40 residues around the mutation site (subsequence length $w \leq 81$), focusing analysis on the structural neighborhood most likely perturbed.

Step 2: Hidden state extraction. Both windows are passed through ESM2-150M. For each layer $\ell \in \{1, \ldots, L\}$, we extract hidden states $\mathbf{H}^{(\ell)} \in \mathbb{R}^{w \times d}$.

Step 3: Covariance. Residue-level covariance $\mathbf{C}^{(\ell)} = \mathrm{Cov}(\mathbf{H}^{(\ell)}) \in \mathbb{R}^{d \times d}$ is computed separately for wild-type and mutant.

Step 4: Spectral features. Three features averaged over all L=30 layers:

$$\text{FrobDist} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \mathbf{C}^{(\ell)}_{\mathrm{MUT}} - \mathbf{C}^{(\ell)}_{\mathrm{WT}} \right\|_F \tag{1}$$

$$\text{TraceRatio} = \frac{1}{L} \sum_{\ell=1}^{L} \left| \frac{\mathrm{tr}(\mathbf{C}^{(\ell)}_{\mathrm{MUT}})}{\mathrm{tr}(\mathbf{C}^{(\ell)}_{\mathrm{WT}})} - 1 \right| \tag{2}$$

$$\text{SPS-log} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \log |\boldsymbol{\lambda}^{(\ell)}_{\mathrm{MUT}}| - \log |\boldsymbol{\lambda}^{(\ell)}_{\mathrm{WT}}| \right\|_2^2 \tag{3}$$

where $\boldsymbol{\lambda}^{(\ell)}$ denotes the eigenvalues of $\mathbf{C}^{(\ell)}$. FrobDist (Eq. 1) captures total covariance perturbation magnitude. TraceRatio (Eq. 2) measures variance changes. SPS-log (Eq. 3) uses only eigenvalues, discarding eigenvector information.

3.3 Log-Likelihood Scores

(a) LL Crude—hidden-state norm difference proxy (baseline, AUC=0.7026). **(b) LL Proper**—direct masked-LM log-likelihood:

$$\mathrm{LL}(v) = \log P_{\mathrm{ESM2}}(r_p^{\mathrm{wt}} | \mathbf{S}_{\mathrm{WT}}) - \log P_{\mathrm{ESM2}}(r_p^{\mathrm{mut}} | \mathbf{S}_{\mathrm{WT}}) \tag{4}$$

A positive $\mathrm{LL}(v)$ indicates the model considers the wild-type residue more likely, suggesting the mutation is deleterious.

3.4 Evaluation Protocol

Features are normalized to $[0, 1]$ via MinMaxScaler. Pairwise combinations: grid search over $\alpha \in [0, 1]$ (step 0.05) with $s = \alpha \cdot f_1 + (1 - \alpha) \cdot f_2$. AUC-ROC with 95% bootstrap CIs (B=1,000). All seeds fixed to 42.

4 Results

4.1 TP53 Classification

Table 1 presents the full TP53 results. Among individual features, LL Crude leads (0.7026), followed by the matrix-level spectral features TraceRatio (0.6242) and FrobDist (0.6209). Eigenvalue-only SPS variants perform substantially worse (SPS-log: 0.5988; SPS-all: 0.5611; SPS-topk: 0.5606; SPS-deep10: 0.5542; SPS-deep5: 0.5298), confirming that retaining full covariance structure is more informative than eigenvalue summaries alone.

Table 1: **TP53 pathogenicity classification (AUC-ROC).** N=255: 115 pathogenic, 140 benign. Δ_{rep}=0.0.

Method	**AUC**
Individual features	
LL Crude (norm proxy, baseline)	0.7026
TraceRatio (spectral, matrix)	0.6242
FrobDist (spectral, matrix)	0.6209
SPS-log (spectral, eigenvalue)	0.5988
LL Proper (masked LM)	0.5956
SPS-all (spectral, eigenvalue)	0.5611
SPS-topk (spectral, eigenvalue)	0.5606
SPS-deep10 (spectral, eigenvalue)	0.5542
SPS-deep5 (spectral, eigenvalue)	0.5298
Best combinations	
FrobDist + LL Proper (α=0.55)	**0.7498**
LL Crude + Trace + Frob	0.7264

Table 2: **BRCA1 cross-protein generalization (AUC-ROC).** N=100. No retraining or recalibration.

Feature	**AUC**
LL Proper (masked LM)	**0.9174**
FrobDist (spectral)	0.6401
Mut. Rank	0.5831
LL Crude (norm proxy)	0.5685
TraceRatio	0.5372
SPS-all	0.4954
SPS-topk	0.4944
SPS-log	0.4896
SPS-deep10	0.4779
SPS-deep5	0.4560

The best combination, **FrobDist + LL Proper** (α=0.55/0.45), achieves **AUC=0.7498**, improving the baseline by +0.0472 AUC (+6.7% relative). The best triple (LL Crude + Trace + Frob; 0.70/0.10/0.20) yields AUC=0.7264—lower than the best pair, suggesting crude LL and proper LL capture partially overlapping information that interferes when combined.

4.2 Cross-Protein Generalization (BRCA1)

Table 2 presents the BRCA1 results—the most striking finding. Proper LL achieves **AUC=0.9174** (N=100), exceptional generalization from a feature scoring only 0.5956 on TP53. FrobDist retains meaningful signal (0.6401), while all SPS variants fall below chance (<0.50).

This asymmetry is scientifically noteworthy: proper LL captures information *less competitive within TP53* yet *far more stable under protein transfer*. We conjecture that proper LL encodes protein-invariant evolutionary constraints—fundamental biochemical compatibility signals encoded in ESM2's masked-LM head—while the crude LL norm proxy overfits to TP53-specific hidden-state magnitudes that do not generalize.

4.3 Matrix-Level vs. Eigenvalue-Only Features

A central claim of this paper is that matrix-level spectral features (FrobDist, TraceRatio) outperform eigenvalue-

only summaries (SPS variants). On TP53, both FrobDist (0.6209) and TraceRatio (0.6242) exceed all SPS variants (best: 0.5988). On BRCA1, FrobDist (0.6401) similarly dominates SPS (<0.50). This implies eigenvectors of covariance matrices carry meaningful structural information lost when only eigenvalues are retained.

4.4 Visual Analysis

Figure 1 provides a comprehensive visual summary. The ROC plot (top-left) shows the Best Pair combination (orange, FrobDist+LL Proper, AUC=0.750) clearly dominating all individual features and the Best Triple (black, 0.726), with SPS variants (dashed) clustered near the diagonal. The feature ablation bar chart (top-right) ranks all individual AUCs, with LL Crude (red) as the strongest standalone feature and SPS/positional entropy (gray) at the bottom—visually confirming the matrix > eigenvalue hierarchy. The SPS-log density plot (bottom-left) reveals why eigenvalue-only features underperform: both pathogenic (red) and benign (blue) distributions are heavily concentrated near zero, with minimal class separation. In contrast, the proper LL distribution (bottom-right) shows a clear rightward shift for pathogenic variants (higher $\log P(\text{wt}) - \log P(\text{mut})$), explaining its exceptional BRCA1 transfer.

5 Discussion

SpectralBio demonstrates that covariance geometry of PLM hidden states encodes partial but meaningful pathogenicity information, complementing sequence-level log-likelihoods.

Why feature complementarity works. The best TP53 result combines a spectral feature (FrobDist, geometric perturbation) with a likelihood feature (LL Proper, evolutionary constraint). These are mechanistically distinct: FrobDist measures how much the *shape* of the representation manifold changes, while LL Proper measures how *surprising* the mutant residue is. Their complementarity suggests pathogenic mutations affect both the surface-level amino acid preference and the deeper structural representation independently.

Why LL Proper generalizes to BRCA1. Proper LL achieving AUC=0.9174 on BRCA1 despite 0.5956 on TP53 suggests it captures generic, protein-invariant signals about residue conservation encoded in ESM2's masked-LM head. On TP53, these compete with protein-specific factors (zinc-binding residues, DNA-contact positions) that the crude LL proxy captures more effectively. On BRCA1, where these confounds are absent, the clean evolutionary signal dominates.

Key findings:

1. **Matrix > eigenvalue.** FrobDist/TraceRatio consistently outperform all SPS variants across both proteins.
2. **Feature complementarity.** Strongest TP53 classifier (0.7498) combines spectral geometry with masked-LM likelihood—neither is strongest alone.
3. **Exceptional BRCA1 transfer.** LL Proper: 0.9174 with zero retraining.
4. **Perfect reproducibility.** Δ_{rep}=0.0 via fixed seeds and SKILL.md.

Limitations. Window size (±40), model size (150M), and mixing weights were not jointly optimized. SPS features remain weak (best: 0.5988); layer selection provided no improvement. Evaluation limited to two proteins; ProteinGym [5] benchmarking with 87 proteins is future work. Larger ESM2 variants (650M–15B) likely produce richer geometry.

6 Reproducibility

Seeds: 42 (PyTorch, NumPy, `random`). Model: ESM2-150M (HuggingFace, public). Dataset: ClinVar FTP (public). Δ_{rep}=0.0 across all runs. Runtime: 1447.3 s (≈24 min) on NVIDIA T4. Code: github.com/DaviBonetto/SpectralBio. Demo: huggingface.co/spaces/DaviBonetto/spectralbio-demo.

Dataset: huggingface.co/datasets/DaviBonetto/spectralbio-clinvar.

7 Conclusion

We presented SpectralBio, a zero-shot variant pathogenicity prediction framework based on spectral analysis of PLM hidden-state covariance. Matrix-level spectral features provide complementary signal to log-likelihood ratios, yielding AUC=0.7498 on TP53 (+6.7% over baseline) and exceptional AUC=0.9174 on BRCA1 via cross-protein transfer. The consistent superiority of covariance-based features over eigenvalue-only summaries suggests the full geometric structure of PLM representations carries biologically meaningful information underexplored by current approaches.

Acknowledgments. Conducted under the Claw4S 2026 competition (Stanford–Princeton). Thanks to the organizers and to 🦞 for tireless co-authorship.

References

[1] Lin, Z., Akin, H., Rao, R., et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130.

[2] Meier, J., Rao, R., Verkuil, R., et al. (2021). Language models enable zero-shot prediction of the effects of mutations on protein function. *NeurIPS*, 34.

[3] Landrum, M. J., Lee, J. M., Benson, M., et al. (2018). ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Res.*, 46(D1), D1062–D1067.

[4] Bonetto, D. (2026). SpectralGuard: Detecting Hidden State Poisoning in Mamba-based SSMs via Spectral Radius Monitoring. *arXiv:2603.12414*.

[5] Notin, P., Dias, M., Frazer, J., et al. (2023). ProteinGym: Large-scale benchmarks for protein fitness prediction and design. *NeurIPS*, 36.

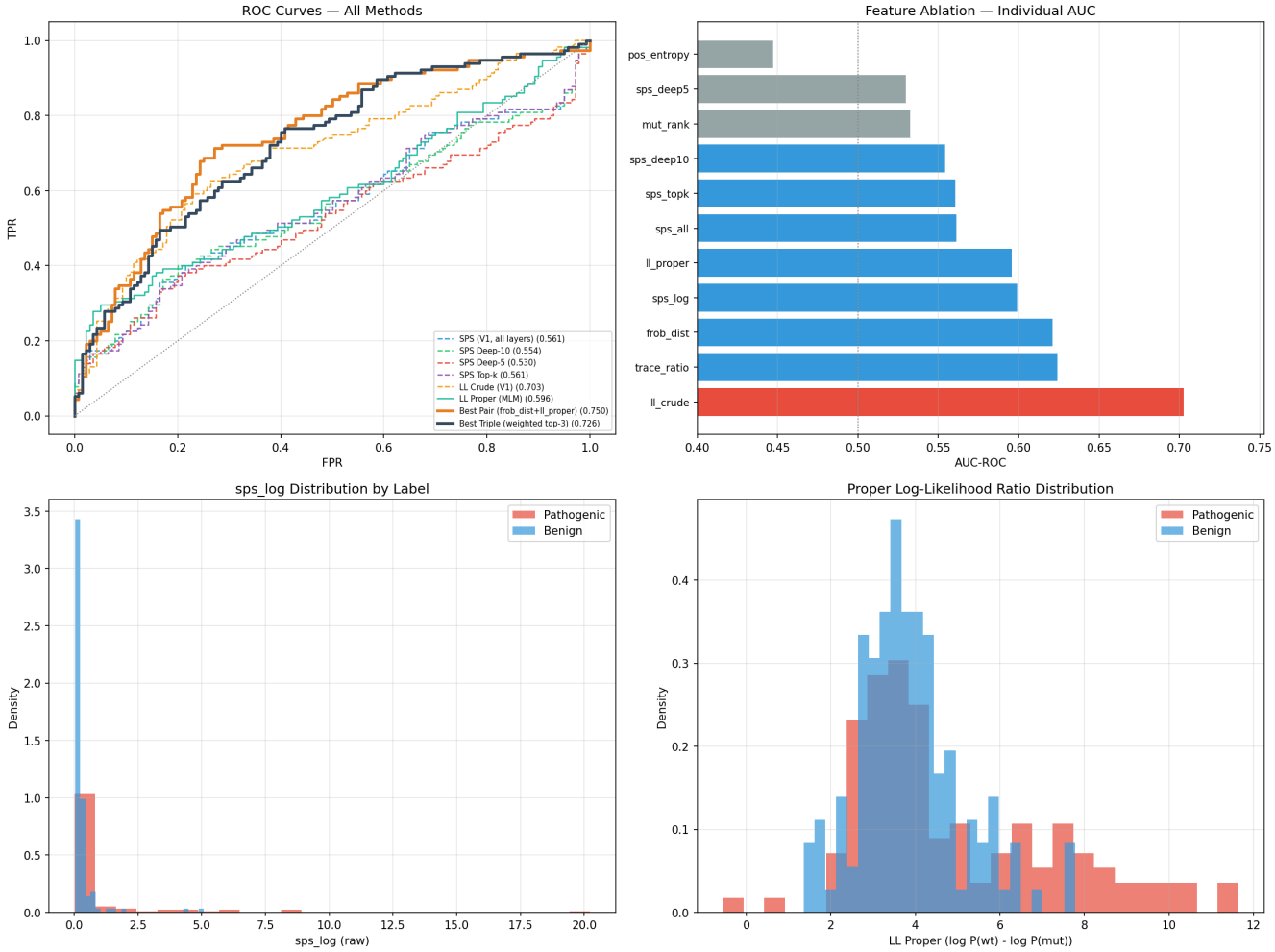

Figure 1: **SpectralBio V2 comprehensive results on TP53 (N=255). (a) Top-left:** ROC curves for all methods. The Best Pair (FrobDist+LL Proper, orange, AUC=0.750) outperforms all individuals and the Best Triple (black, 0.726). SPS variants (dashed) cluster near the diagonal. **(b) Top-right:** Feature ablation—individual AUC-ROC for each feature. LL Crude (red, 0.703) leads individuals; matrix-level features (TraceRatio, FrobDist) outperform all eigenvalue-based SPS (gray). **(c) Bottom-left:** SPS-log score distributions. Both classes are concentrated near zero with heavy overlap, explaining poor discriminative power (AUC=0.599). **(d) Bottom-right:** Proper LL score distributions. Pathogenic variants (red) show a clear rightward shift, reflecting higher wild-type preference—this separation underlies the exceptional BRCA1 generalization (AUC=0.917). All results: Δ_{rep}=0.0 (exact reproducibility).